

CHANGTING, China

WMO: 589110

Lat: 25.8527N

Lon: 116.3659E

Elev: 1020

StdP: 14.16

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.9	33.7	20.6	16.0	42.8	25.0	19.7	41.5	13.7	45.3	12.1	46.6	1.9	320	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	94.3	77.2	92.6	77.0	90.9	76.6	79.4	88.6	78.7	87.8	78.0	86.7	5.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	147.4	82.7	76.5	143.4	82.1	75.8	140.1	81.5	43.9	88.5	43.1	87.9	42.4	87.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.0	10.2	8.8	DB	26.2	97.3	2.8	1.4	24.2	98.3	22.6	99.2	21.0	99.9	18.9	101.0
			WB	25.0	81.0	2.9	1.1	22.8	81.8	21.1	82.5	19.5	83.1	17.3	84.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.2	48.5	52.7	58.8	67.8	74.3	79.1	82.4	81.8	77.8	70.2	61.2	51.4
	DBStd	13.23	7.69	9.23	7.63	6.22	4.46	3.45	2.46	2.71	4.54	5.28	6.78	6.80
	HDD50	283	124	74	15	1	0	0	0	0	0	6	63	
	HDD65	1719	512	354	220	44	2	0	0	0	15	148	424	
	CDD50	6577	78	149	286	536	753	873	1005	987	834	627	342	107
	CDD65	2540	1	10	27	129	290	423	540	522	384	177	34	3
	CDH74	21854	4	63	149	702	1902	3552	5695	5155	3217	1184	223	8
	CDH80	8105	0	5	12	139	558	1225	2500	2119	1223	296	28	0
Wind	WSAvg	3.2	3.5	3.4	3.1	3.1	2.8	2.8	3.1	3.1	3.2	3.3	3.2	3.4
Precipitation	PrecAvg	68.3	2.3	3.9	7.5	9.1	11.2	12.0	4.7	5.9	4.2	2.4	1.8	1.7
	PrecMax	96.5	6.9	11.1	15.7	22.1	20.6	26.4	10.9	14.6	10.0	7.6	6.3	6.6
	PrecMin	49.0	0.1	0.0	1.2	2.6	2.9	0.6	0.5	1.1	0.3	0.0	0.0	0.0
	PrecStd	11.3	2.2	2.4	3.9	4.3	3.6	6.0	2.4	3.3	2.5	2.0	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.8	80.1	81.8	87.1	90.9	93.5	96.9	96.1	94.0	89.5	83.4	74.8
		MCWB	61.6	66.0	67.1	70.6	72.7	77.5	76.4	77.2	77.0	73.7	67.8	64.5
	2%	DB	69.5	75.6	78.3	83.9	88.3	91.3	94.5	94.2	91.8	85.9	79.7	70.3
		MCWB	59.4	63.7	65.9	70.4	72.7	76.5	77.1	77.4	76.4	70.9	66.2	59.9
	5%	DB	65.4	71.8	74.8	81.2	85.8	89.2	92.8	92.2	89.6	83.3	76.3	66.6
		MCWB	56.6	62.1	63.9	69.0	72.6	76.5	77.1	77.4	75.8	69.3	63.9	57.1
	10%	DB	61.2	67.0	71.0	78.3	83.2	86.9	90.9	89.9	87.0	80.7	72.8	62.9
		MCWB	54.9	59.5	63.0	68.2	72.5	76.1	76.9	77.1	74.7	68.1	62.4	55.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.3	68.6	70.3	74.2	78.3	80.0	80.3	80.3	79.4	75.8	71.7	67.6
		MCDB	70.4	76.6	77.9	81.4	85.1	88.2	89.1	90.8	89.4	85.2	79.5	72.1
	2%	WB	61.4	65.6	68.4	72.7	76.8	78.7	79.2	79.2	78.0	74.0	69.3	62.8
		MCDB	67.1	73.1	74.7	79.7	83.5	86.4	88.4	89.1	87.6	81.4	75.5	67.1
	5%	WB	58.6	63.2	66.5	71.5	75.6	77.9	78.5	78.5	77.1	72.5	67.2	60.0
		MCDB	64.3	69.8	71.9	78.0	81.5	85.3	87.8	87.8	86.3	79.4	72.6	64.1
	10%	WB	55.6	60.6	64.5	70.3	74.4	77.1	77.8	77.8	76.1	71.0	65.1	57.1
		MCDB	60.2	66.3	69.3	75.9	79.7	84.3	86.8	86.5	84.2	77.5	70.1	61.6
Mean Daily Temperature Range	5% DB	MDBR	14.9	14.7	14.4	14.5	14.1	12.9	16.0	15.0	15.6	17.7	18.1	17.6
		MCDBR	23.8	22.9	22.2	20.5	20.5	17.7	19.9	19.5	19.4	22.3	24.4	24.9
		MCWBR	13.2	11.5	10.1	8.3	7.3	5.4	5.2	5.4	5.8	8.4	10.9	13.5
	5% WB	MCDBR	19.0	19.4	16.3	14.8	13.5	14.0	16.6	16.5	16.6	16.2	16.5	16.5
		MCWBR	11.1	10.5	9.1	7.1	5.7	5.2	5.4	5.5	5.7	6.8	8.5	10.3
Clear-Sky Solar Irradiance	taub	0.604	0.659	0.761	0.739	0.622	0.583	0.556	0.612	0.618	0.628	0.570	0.554	
	taud	1.668	1.606	1.464	1.548	1.815	1.923	1.998	1.834	1.792	1.708	1.789	1.781	
	Ebn at Noon	200	202	191	202	227	234	240	227	220	207	209	207	
	Edh at Noon	69	79	96	90	69	61	57	67	68	70	61	60	
All-Sky Solar Radiation	RadAvg	759	834	904	1061	1194	1276	1694	1529	1352	1198	934	852	
	RadStd	159	176	148	168	183	145	151	131	127	199	142	171	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.76	N/A	N/A	+1.35	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.58	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	+214	+157

Nomenclature: See separate page